TOBE. ENERGY

The Biggest Drop in Green Hydrogen Costs. Ever.

The better fuel *finally* has better prices.

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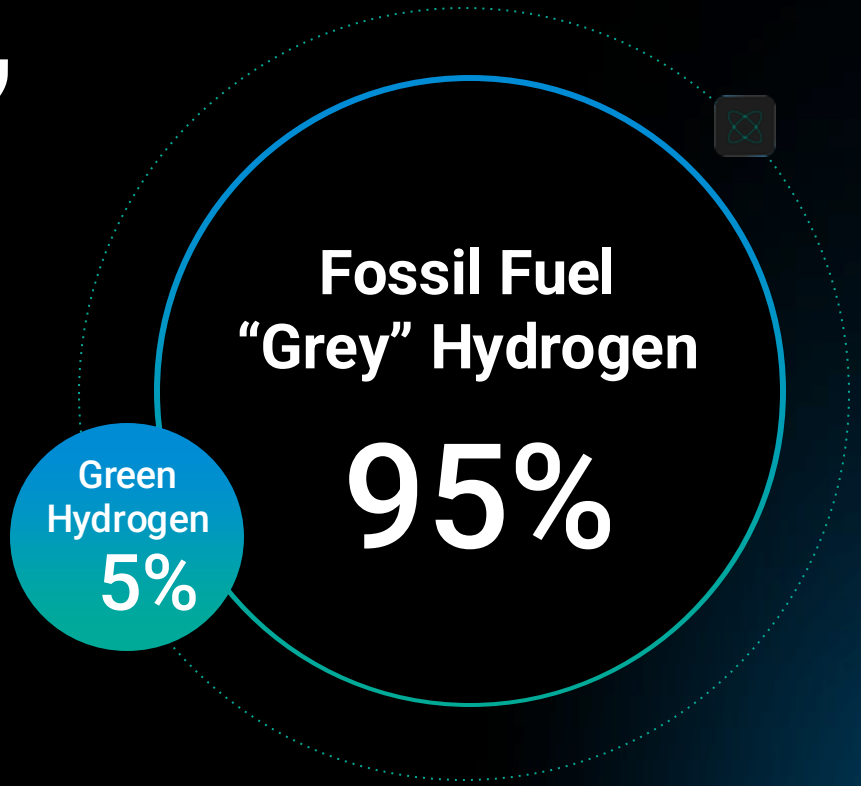
Without us, hydrogen runs on fossil fuels.



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Companies want to go green.

Costs keep them stuck.

77%

of executives are unwilling to pay a premium for green technologies

Capgemini

85%

of organizations are concerned about reducing their greenhouse gas emissions

BCG

CO_2

Fossil fuel derived hydrogen emits more CO2 than the entire aviation industry

iea

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Fertilizer. Fuel. Food. Pharma.

Hydrogen is already everywhere.

CURRENT	FUTURE
Fertilizer	Aviation
Food	Transport
Medical	Heavy Industry
Pharma	Grid Backup
Metal	Industrial Heat
Refining	AI Data Centers
$150B	$1T+

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While the market *hopes* for a dramatic price decrease...

Competitors are still using 100+ year-old methods.

Think about what happened to lighting with LEDs.

Green hydrogen *needs* paradigm-shifting tech.

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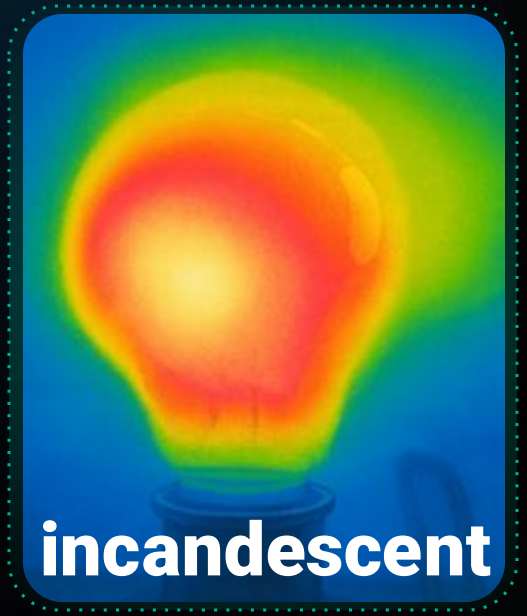
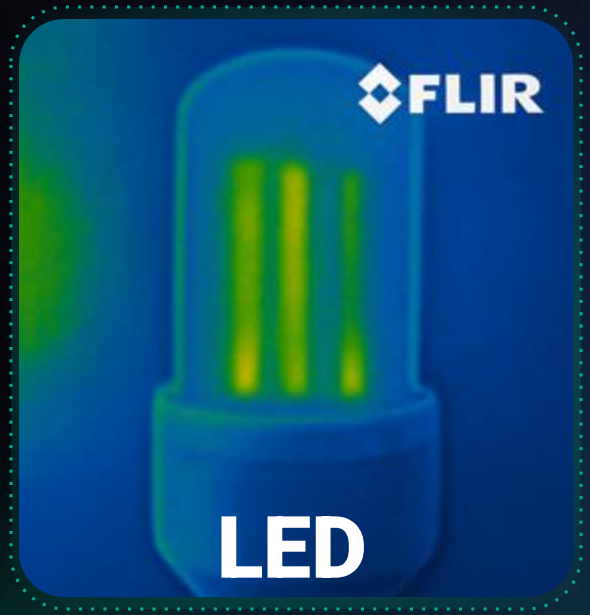


>90%

energy lost as **heat**



5X

Cheaper to run

85%

Market **share**

So we built it.

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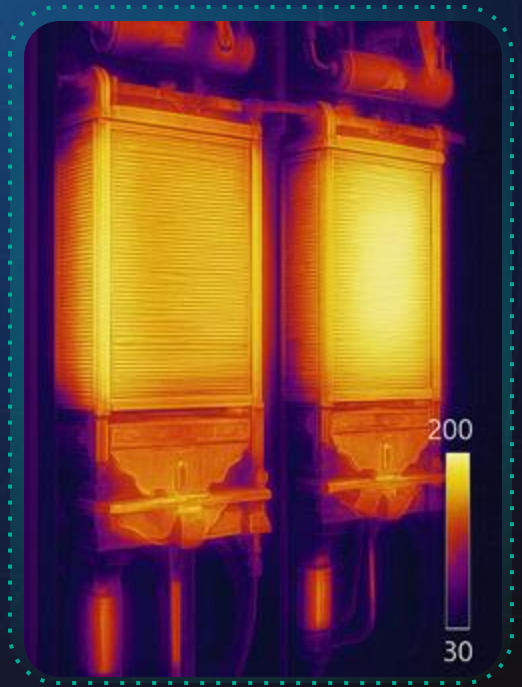
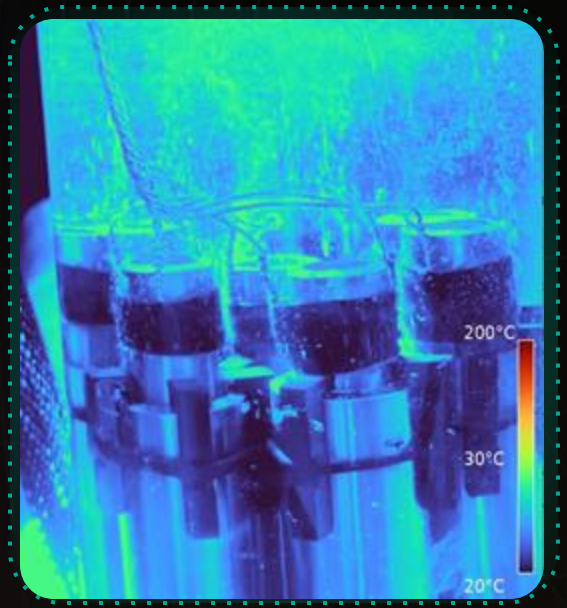


>40%
energy lost
as **heat**

$40 for every
$100 of
electricity used
is wasted



8X
Less
waste

<5%
energy lost
as **heat**

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A change this big is going to be disruptive…

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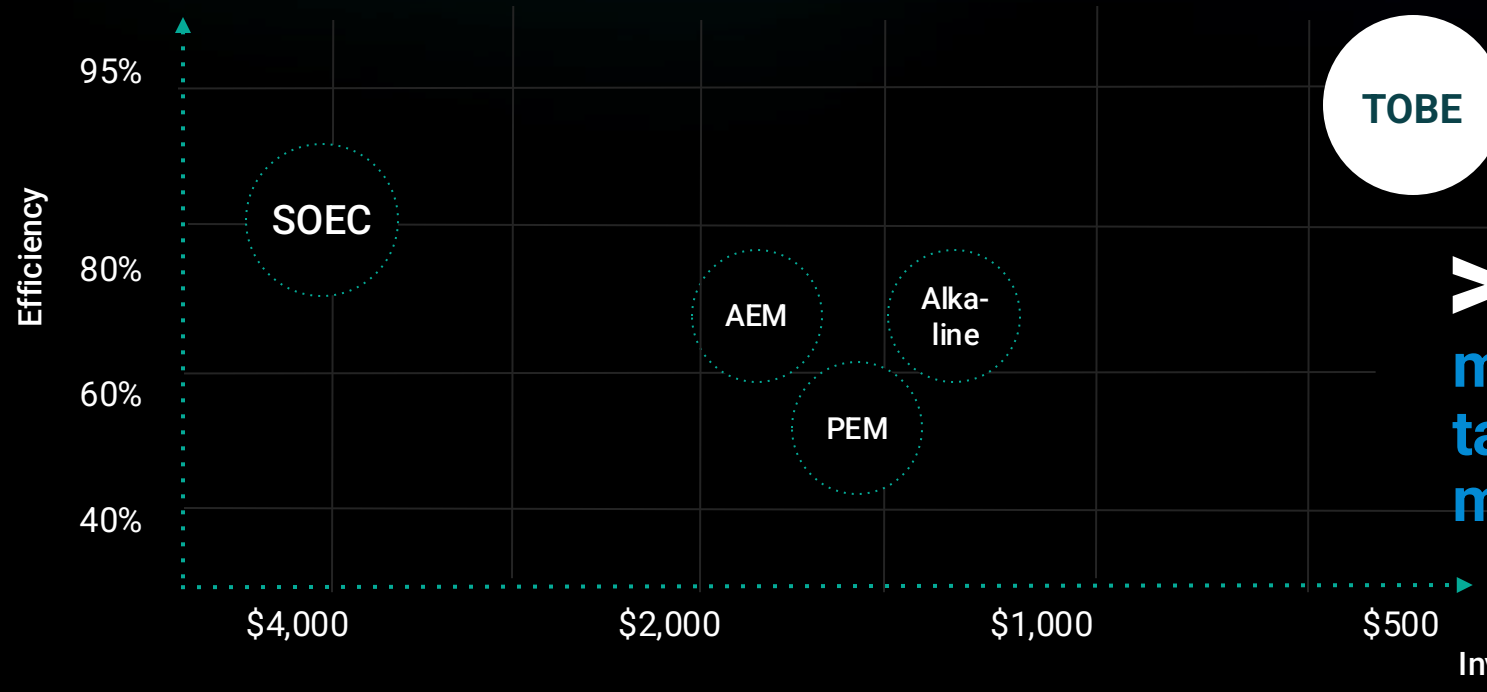

>80%
margin in targeted markets

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Tobe hydrogen replaces batteries in AI Data Centers.

Lower cost & no rare earth materials.

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Massively decreasing alternative fuel costs.

Overnight.

e-Fuels

35%

$10 / **gal** -> **$6.50** / gal

SAF

35%

$15 / gal -> **$9.75** / gal

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Even green fertilizer benefits.

42.5%

$1,000 / ton -> **$575 / ton**

Enabling carbon negative farming.

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Our cheap green H_2 will expand a $150B industry into a $1T+ industry.

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We had to completely re-imagine electrolysis to make it happen.

Power Electronics

No waste heat.
2x the efficiency.
No one else has it (IP Protected).

Built to last materials

All-American steel.
No rare earth metals.
No costly replacements.
Lower costs with longer life.

Re-Industrialize

Vertically integrated American-made manufacturing.
Margin-protecting.
Moat-securing.

Comparison of non-conventional Electrolysis Startups.

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Category	TOBE. ENERGY	Advanced Ionics	Fourier
Electrolysis Type	✓ Resonant, low-temp	⚠ RSOEC-like vapor-phase	✗ PEM stack (conventional)
Rare Earths	✓ None (Stainless Steel)	⚠ Uses Nickel & Stainless	✗ Platinum, IrO_2
High Voltage	✓ Yes	✗ No	✓ Yes
Membraneless	✓ Yes	✓ Yes (vapor-phase)	✗ No (PEM membrane)
Efficiency (HHV basis)	● 94+% (~42 kWh/kg)	● 35 kWh/kg (target only) (~75% HHV est. incl. waste heat)*	● 50–55%
Engineering In-House	✓ Full-stack: system + cell + circuit	⚠ Stack only	✗ Off-the-shelf PEM
Manufacturing In-House	✓ Yes (full proto capability)	✗ No (partnered scale-up)	✗ No (assembly only)
Pilot Size	● Mid (6,400 cm² / 25 kg/day)	● Small (800 cm²)	● Small (modular blade)
Pilot Cost	● $300K (self-built, low-cost)	● $4M (thermal + integration)	● $5M (modular PEM)
Operating Temp	● <30°C (low-temp)	● ~300°C (high-temp)	⚠ Unknown (likely high)

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Sell hydrogen. Build infrastructure. Scale.

Hydrogen-as-a-Service (HaaS)

Build local delivery networks to slash delivery costs and improve speed.

Expand Site-by-Site

Starting with the highest margin states first.

Scale Fast

Lease tech, license IP, expand to ammonia, SAF, and AI power

Economics of Small, Modular, Local Production

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$4 / kg
opex includes personnel & site

$5 M
capex / site includes electrical expansion

900 t / yr
produced hydrogen

Cost Line Item	Long-Haul	Tobe
Production	$4 / kg	$4 / kg
Liquify & Cryo Trailer (300 - 500 mi)	+ $7	
Vaporize & Recompress	+ $3	
Short Haul Trailer	+ $1	+ $1
Delivered Cost	≈ $15	≈ $5

$10 M annual profit / site

63% net margins

~12 mo. payback after production begins

8,500 t / yr of CO2 saved

One $5 M Tobe site reduces over 8,500 tonnes of CO_2 per year—roughly the footprint of 2,000 cars—while producing $10–12M in cash.

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In *just* 10 weeks...

$20M

in signed demand

Timeline & Use of Funds

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Pre-seed
Expand mfg ops
Hire key talent

Seed
Co-dev of first operational site with Zeeco

Series A
Expand into 2 additional high margin markets

Build 25 kg / day units
R&D 125 kg / day

Build & Run
Built on 25 kg units

Deliver on LOIs
Provide on site hydrogen @ Zeeco

The faster we close the Seed, the faster we lock in 10 years of §45V tax credits (worth $1M+) and flip $20 M of contracted demand into real cashflow—while Zeeco's campus gives us a de-risked proving ground for every module we'll cookie-cut across the country.

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Team

$75M+ Projects Managed

Greenfield Natural Gas Facility, Oil Terminal, 10 years in the Energy Industry.

Delivered FOAK Tech

Brought first-of-its-kind zero-emission hydrogen boiler to market. Led engineering, procurement & design.

Colby DeWeese & Dr. Caleb Lareau

Founding Team:

Stanford ARKEMA SC MARATHON HARVARD

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Clean hydrogen has always been the future. But starting today, **it's the present**.

Colby DeWeese
colby@tobe.energy